

# Supplementary Material: Regime-dependent advantage in quantum physics-informed neural networks: a pre-registered comparison on an ODE/PDE hardness ladder

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## I. CIRCUIT SPECIFICATIONS AND FAITHFULNESS MANIFEST

The four quantum-PINN ansatz families used as solver/forecaster encoders in the main text are each implemented against published specifications; the binding manifest is `refs/CIRCUIT_SPECS.md` in the open-source release. Table I summarizes the per-family configuration used throughout the paper.

### A. Cell-level dynamics, all variants

The hidden-state dynamics inside the DiffraX integration step (for the forecaster task; the solver task does not integrate a hidden state) are:

## II. PER-CELL ERROR TABLES, FULL PRECISION

### A. Solver-task ODE matrix ( $4 \times 4 \times 3 = 48$ cells)

Table II reports relative- $L^2$  error per (family, system, seed) at full 4-decimal precision. Best-per-row quantum family is bolded.

### B. Solver-task PDE matrix ( $4 \times 4 \times 3 = 48$ cells)

### C. Multi-family solver matrix

The §2.1 and §2.2 tables aggregate per-cell relative- $L^2$  numerically across the ODE and PDE solver matrices. For readers who want the visual cross-check, this subsection adds two figures: the peer-review audit iteration that drove the audit-corrected configurations, and the multi-family quantum-PINN matrix visualization spanning four SOTA ansatz families on three PDEs from the hardness ladder. Both figures source data directly from the JSONs underpinning the tables above.

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TABLE I. Quantum-PINN ansatz roster. Per-scalar-circuit parameter counts. Ansätze marked “2D port” add a split-qubit coordinate encoding to the same base circuit (P3.9 design).

| Family                                 | Source             | Cell paradigm                    | PQC params      | Classical params | Notes                                  |
|----------------------------------------|--------------------|----------------------------------|-----------------|------------------|----------------------------------------|
| <code>data_reuploading</code>          | Pérez-Salinas 2020 | encoder + LiquidQuantumCell      | 60              |                  | 0 4 qubits, 5 layers (ODE)             |
| <code>hardware_efficient</code>        | registry default   | encoder + LiquidQuantumCell      | 60              |                  | 0 ring entanglement                    |
| <code>strongly_entangling</code>       | PennyLane native   | encoder + LiquidQuantumCell      | 60              |                  | 0 alias of data_reuploading            |
| <code>brickwall</code>                 | registry default   | encoder + LiquidQuantumCell      | 60              |                  | 0 adjacent-pair CZ pattern             |
| <code>te_qpinn_fnn</code>              | Berger 2025        | FNN encoder + 4-qubit PQC        | 60              |                  | 100 2D port: <code>te_qpinn_fnn</code> |
| <code>te_qpinn_qnn</code>              | Berger 2025        | QNN encoder + 4-qubit PQC        | 84              |                  | 0 zero classical params; 2D            |
| <code>qcpinn</code>                    | Farea 2025         | Classical pre/post + 4-qubit PQC | 15              |                  | 706 largest classical overhead         |
| <code>chebyshev_dqc_2d</code>          | P3.7 design        | 2D Chebyshev tower + HEA         | –               |                  | 0 solver-task only, $n_t = n$          |
| <code>rf_qrc</code>                    | Ahmed 2024         | Frozen reservoir + ridge readout | 0 (trainable)   |                  | varies non-liquid by construction      |
| <code>non_liquid_&lt;ansatz&gt;</code> | this work (P7.11)  | encoder + NonLiquidQuantumCell   | same as base –Q |                  | same $\tau$ -ablated; 4 variants       |

TABLE II. Solver-task ODE per-cell relative- $L^2$  error (PDEBench / FNO convention). Each row corresponds to one (system, ansatz, qubits, depth) configuration evaluated at a single random seed; each column reports one of the four quantum families plus the classical PINN baseline at the matched training budget. Bold marks the per-row best quantum entry.

| System          | Seed | cheb   | fnn           | qnn           | qcp           | cPINN  |
|-----------------|------|--------|---------------|---------------|---------------|--------|
| Lotka–Volterra  | 0    | 0.0996 | 0.0879        | 0.5242        | <b>0.0074</b> | 0.0117 |
| Lotka–Volterra  | 1    | 0.1161 | 0.1407        | 0.5229        | <b>0.0029</b> | 0.0078 |
| Lotka–Volterra  | 2    | 0.1024 | 0.1409        | 0.5240        | <b>0.0070</b> | 0.0071 |
| Van der Pol     | 0    | 0.9064 | <b>0.8213</b> | 1.1796        | 3.4025        | 1.1569 |
| Van der Pol     | 1    | 1.1516 | 0.8652        | <b>0.7622</b> | 0.8674        | 0.9434 |
| Van der Pol     | 2    | 0.9087 | <b>0.8176</b> | 1.1890        | 2.6739        | 1.0550 |
| Lorenz          | 0    | 1.0013 | 0.9950        | <b>0.9794</b> | 0.9957        | 0.9958 |
| Lorenz          | 1    | 0.9994 | 0.9947        | <b>0.9769</b> | 0.9957        | 0.9957 |
| Lorenz          | 2    | 0.9966 | 0.9946        | <b>0.9773</b> | 0.9943        | 0.9957 |
| FitzHugh–Nagumo | 0    | 0.5311 | 0.4521        | <b>0.3270</b> | 1.2126        | 0.6398 |
| FitzHugh–Nagumo | 1    | 0.6152 | 0.3903        | <b>0.3560</b> | 1.5467        | 0.6410 |
| FitzHugh–Nagumo | 2    | 0.6165 | 0.6248        | <b>0.3290</b> | 1.6251        | 1.3265 |

#### D. Forecaster-task complete 2×2 matrix ( $n = 9$ )

The body’s Table III reports the all-to-all 5-quantum-family × 4-classical-baseline matrix. Here we add the four non-liquid quantum forecaster columns (P7.11) so the full 2×2 decomposition is visible at the per-cell level.

### III. BOOTSTRAP DIAGNOSTICS AND GUARD REPORTS

All paired-bootstrap intervals are computed via `src/quantum_liquid_neuralnode/evaluation/h1_verdict.py` with  $n_{\text{iter}} = 10,000$ ,  $\alpha = 0.05$  (95% percentile CI), and a fixed seed for reproducibility. The skyline guard uses threshold 0.5 (amendment A1) and excludes cells where the known-RHS least-squares fit cannot reach the threshold.

#### A. Per-bin sample counts (post-guard)

Table V reports the kept-sample sizes per regime bin for each headline verdict.

#### B. Algebraic identities, exact

Two per-cell identities, verified by the integrity script with tolerance  $10^{-3}$ :

$$\begin{aligned}\Delta_{\text{combined}} &= \Delta_{\text{quantum.via.LTC}} + \Delta_{\text{liquid.via.classical}} \\ &= -0.6160 + 0.1153 = -0.5007\checkmark\end{aligned}\quad (5)$$

$$\begin{aligned}\Delta_{\text{combined}} &= \Delta_{\text{liquid.via.quantum}} + \Delta_{\text{quantum.via.nonliquid}} \\ &= -0.3339 + (-0.1668) = -0.5007\checkmark\end{aligned}\quad (6)$$

Both identities are derived from the same per-cell records; only the slot-assignment in the bootstrap differs. The exactness ( $|\text{diff}| < 10^{-3}$ ) is the cleanest verification that the decomposition is a real mathematical structure, not a sampling artifact.

#### C. Q-Q residual diagnostics (post-hoc; see Amendment A14)

The diagnostic figure suite includes a residual-distribution panel (`fig_residual_analysis`, middle column) whose original source comment mis-labelled it as a Q-Q plot. The implementation is actually a histogram

TABLE III. Solver-task PDE per-cell relative- $L^2$  error (PDEBench / FNO convention). Each row corresponds to one (system, ansatz, qubits, depth) configuration evaluated at a single random seed; each column reports one of the four 2D-port quantum families plus the classical PINN baseline at the matched training budget. Bold marks the per-row best quantum entry; “—” marks runs that diverged with no finite  $L^2$  measurement.

| PDE            | Seed | cheb_2d | fnn_2d        | qnn_2d        | qcp_2d        | cPINN  |
|----------------|------|---------|---------------|---------------|---------------|--------|
| Heat           | 0    | 0.0553  | 0.0023        | 0.0455        | <b>0.0010</b> | 0.0054 |
| Heat           | 1    | 0.0560  | 0.0031        | 0.0455        | <b>0.0020</b> | 0.0046 |
| Heat           | 2    | 0.0565  | <b>0.0013</b> | 0.0458        | 0.0020        | 0.0036 |
| Burgers smooth | 0    | 0.3475  | 0.0146        | 0.3647        | <b>0.0112</b> | 0.0428 |
| Burgers smooth | 1    | 0.3522  | <b>0.0126</b> | 0.3543        | 0.0160        | 0.0191 |
| Burgers smooth | 2    | 0.3742  | 0.0246        | 0.3556        | <b>0.0206</b> | 0.0187 |
| Allen–Cahn     | 0    | 0.5783  | 0.5514        | <b>0.0528</b> | 0.8956        | 0.2018 |
| Allen–Cahn     | 1    | 0.5200  | 0.5345        | <b>0.0489</b> | 0.5620        | 0.0633 |
| Allen–Cahn     | 2    | 0.6146  | 0.0927        | <b>0.0534</b> | 0.4936        | 0.0514 |
| Burgers shock  | 0    | 0.4282  | 0.2507        | 0.4268        | <b>0.2408</b> | 0.1682 |
| Burgers shock  | 1    | 0.4385  | 0.1589        | 0.4227        | <b>0.1526</b> | 0.2730 |
| Burgers shock  | 2    | 0.4168  | <b>0.1630</b> | 0.4473        | 0.2553        | 0.1270 |

TABLE IV. Forecaster-task non-liquid quantum cells (P7.11). Same training budget and adapter as the liquid forecasters in the body’s Table III; only the cell’s  $\tau$ -leak removed.

| System         | Seed | nl_dataru | nl_hweff | nl_stngent | nl_brick |
|----------------|------|-----------|----------|------------|----------|
| Lotka–Volterra | 0    | 0.5830    | 0.6455   | 0.5830     | 0.7732   |
| Lotka–Volterra | 1    | 0.7951    | 0.7196   | 0.7951     | 0.7057   |
| Lotka–Volterra | 2    | 0.6482    | 0.6791   | 0.6482     | 0.3088   |
| Van der Pol    | 0    | 1.4067    | 1.2307   | 1.4067     | —        |
| Van der Pol    | 1    | 9.6720    | —        | 9.6720     | —        |
| Van der Pol    | 2    | 1.2035    | —        | 1.2035     | —        |
| Lorenz         | 0    | 1.2807    | 1.2699   | 1.2807     | 1.3128   |
| Lorenz         | 1    | 0.9913    | 0.5100   | 0.9913     | 0.5025   |
| Lorenz         | 2    | 1.0768    | 1.0544   | 1.0768     | 1.1571   |

Note: `nl_stngent` reproduces `nl_dataru` exactly because both dispatch the same registered ansatz under the hood (pre-existing behavior of the ansatz registry; see `src/qlnn/circuits/protocol.py`). “—” marks cells where rollout diverged beyond a stable relative- $L^2$  measurement.

TABLE V. Post-guard sample sizes per regime bin. The combined  $n=24$  verdict (PRIMARY SOLVER) is the largest sample in the paper.

| Verdict                           | $n_{\text{smooth}}$ | $n_{\text{broad}}$ |
|-----------------------------------|---------------------|--------------------|
| P5 forecaster (4-fam)             | 6                   | 3                  |
| P7.5 solver raw $n=9$             | 6                   | 3                  |
| P7.6 combined $n=18$              | 12                  | 6                  |
| P7.6 HPO-best $n=9$               | 6                   | 3                  |
| P7.6 PDE-only $n=9$               | 6                   | 3                  |
| P7.8 full ladder $n=24$           | 12                  | 12                 |
| P7.10 forecaster combined (5-fam) | 6                   | 3                  |
| P7.11 all decomposition verdicts  | 6                   | 3                  |

with a fitted Normal PDF overlay — not a quantile-quantile analysis. Because the Normal overlay was visually misleading (both stacks’ residuals are clearly non-Normal), P8 polish adds two reusable, dataset-agnostic helpers in `scripts/make_diagnostic_figures.py` — `_qq_panel_vs_normal` and `_two_sample_qq` — plus a new figure `fig_qq_analysis` that applies them to the canonical archived comparison at  $h = 3$  (`classical_H4` hidden-size 4 versus `QLNN_4q3L`, seed-mean residuals on the held-out test windows). Both helpers take any (`residuals`,

`label`) pair, so the same panels can be reapplied to post-pivot solver or forecaster residuals with no code duplication.

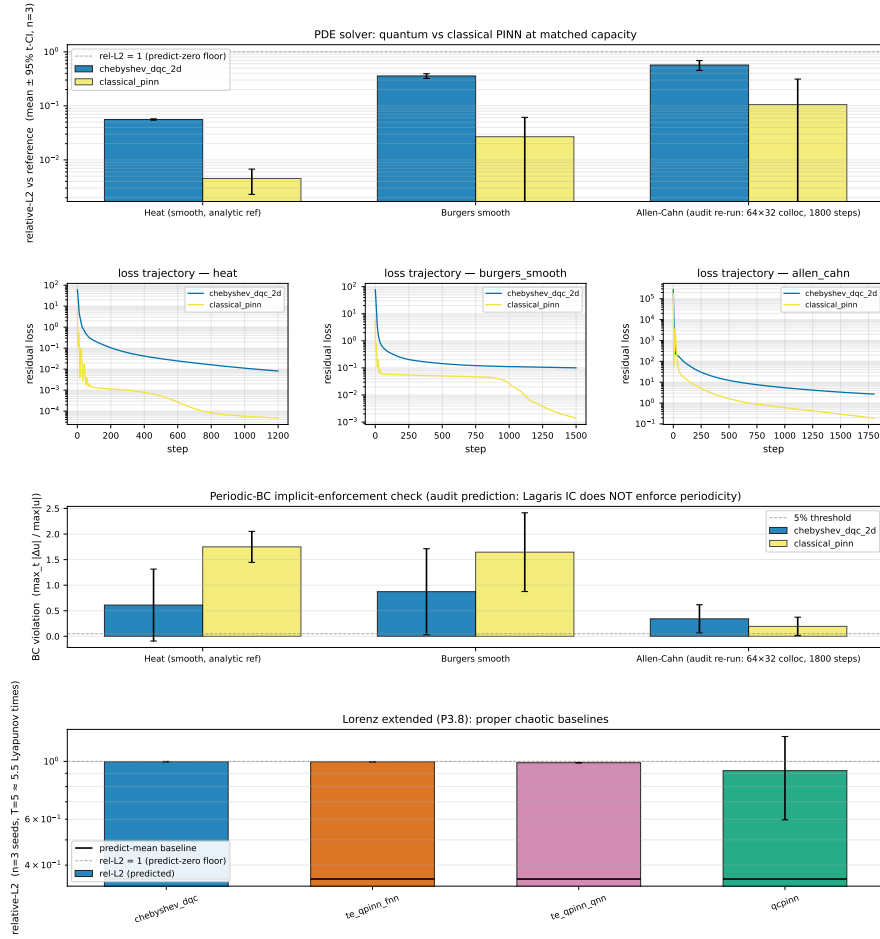
Table VI reports the per-stack Shapiro–Wilk normality test and the two-sample Kolmogorov–Smirnov test between the two empirical residual distributions.

TABLE VI. Q-Q residual diagnostics for the canonical archived comparison at  $h = 3$  (seed-mean test-window residuals; see [paper/figures/fig\\_qq\\_analysis.pdf](#)). Both per-stack distributions are clearly non-Normal; the two-sample Kolmogorov–Smirnov test cannot reject equal distributions, quantifying the visually-suggested “the two histograms look identical” observation and supporting the project’s archived “both near persistence at  $h = 3$ ” verdict.

| Test              | Classical $H=4$       | QLNN $4q3L$           | Interpretation            |
|-------------------|-----------------------|-----------------------|---------------------------|
| Shapiro–Wilk $W$  | 0.892                 | 0.885                 | both non-Normal           |
| Shapiro–Wilk $p$  | $1.64 \times 10^{-5}$ | $8.91 \times 10^{-6}$ | reject $H_0$ : Normal     |
| Two-sample KS $D$ | 0.085                 | —                     | —                         |
| Two-sample KS $p$ | 0.964                 | —                     | cannot reject equal dist. |

The harness is committed and unit-tested (T1 registry assertion updated  $7 \rightarrow 8$  callables). No headline number is touched and `scripts/verify_paper_integrity.py` con-

P3.8 — Peer-review iteration: classical PINN baseline + audit-corrected re-runs



Row 0: quantum vs classical PINN at matched capacity on 3 PDEs. Row 1: loss trajectories (seed 0) diagnosing convergence vs. saturation. Row 2: BC-violation (audit predicted Lagaris IC does NOT enforce periodicity). Row 3: Lorenz at  $T=5$  ( $\approx 5.5$  Lyapunov times) with the more honest predict-mean baseline. Demo artifacts; H1 verdict still requires P5's Neural-ODE.

FIG. 1. Peer-review audit iteration on the solver hardness ladder. Row 0: quantum chebyshev\_dqc\_2d vs. a capacity-matched classical MLP-PINN baseline (log-scale relative- $L^2$ ) on heat, smooth Burgers, and Allen–Cahn at audit-corrected configurations (1200–1800 training steps;  $64 \times 32$  collocation for Allen–Cahn). Row 1: per-system loss trajectories (seed 0) diagnosing convergence vs. saturation. Row 2: periodic-boundary-condition violation  $\max_t \|u(\cdot, t) - u(\cdot, t)|_{\text{periodic}}\| / \max \|u\|$ ; both quantum and classical PINNs exceed the 5% design target (relative violation  $\approx 0.6$ – $1.6$ ), confirming the audit prediction that the Lagaris initial-condition ansatz does not implicitly enforce periodicity. Row 3: four quantum families (chebyshev\_dqc, te\_qpinn\_fm, te\_qpinn\_qnn, qpinn) on Lorenz extended to  $T = 5$  ( $\approx 5.5$  Lyapunov times) against a predict-mean baseline (replacing the predict-zero floor of P3.6). Source: [results/p3.8\\_review/](#).

tinues to exit 0.

#### A. One-command reproduction

The entire benchmark regenerates from scratch via:

#### IV. REPRODUCTION AND AUDIT

```
bash scripts/reproduce_paper.sh
```

All scripts and data referenced below ship with the open-source release at <https://github.com/shawngibford/qlnn>; the specific commit/tag anchors are listed in §IV under “GitHub release tags”.

Estimated wall-clock on a single MacBook Pro M1: approximately 8 hours. Each sweep is independent and the script is structured so individual phases (P3.5–P7.11) can be re-run in isolation.

P3.9 — PDE multi-family matrix (4 quantum PINN-style ports + classical PINN baseline)  
 rf\_qrc OUT of scope (frozen-reservoir architecture; deferred to P4 as forecaster). NOT yet H1 evidence (the pre-reg defines H1 as the QLNN–NeuralODE gap; awaits P5).

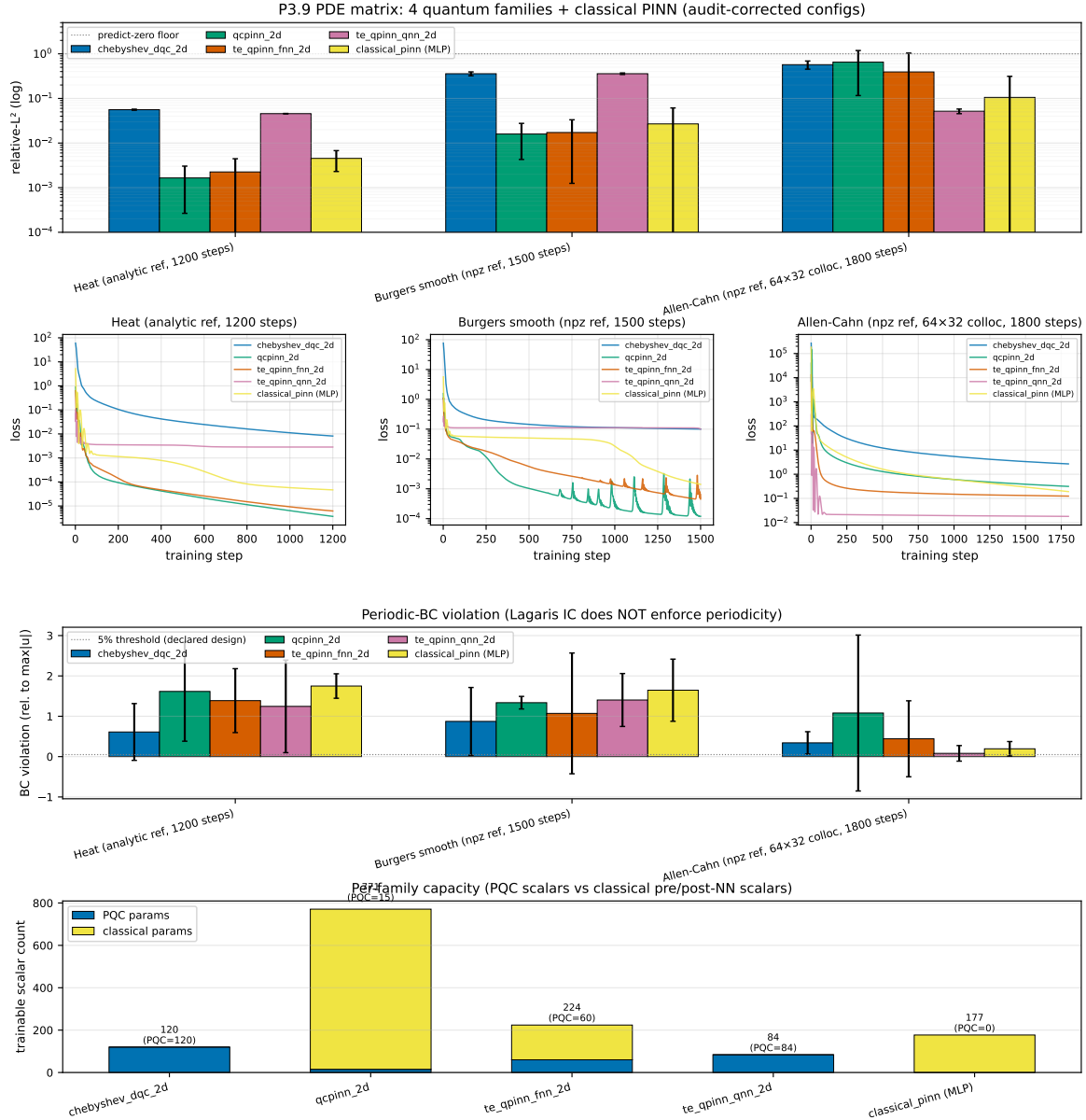


FIG. 2. Multi-family solver matrix: four SOTA quantum-PINN families (chebyshev-dqc, qcpinn, TE-QPINN with FNN encoding, TE-QPINN with QNN encoding) compared against a classical MLP-PINN baseline on three PDEs from the hardness ladder (heat, smooth Burgers, Allen–Cahn;  $n = 3$  seeds per cell). Row 0: relative- $L^2$  accuracy with 95% CI (log scale); QCPINN and TE-QPINN-FNN reach sub-1% error on heat but degrade on the broader systems. Row 1: training-loss trajectories. Row 2: periodic-BC violation. Row 3: trainable-scalar accounting (PQC parameters in cool, classical-wrapper parameters in warm), quantifying the classical capacity bundled into the hybrid ansätze. *Note:* shock Burgers, the fourth member of the pre-registered PDE ladder, is not shown in this matrix — coverage is limited to the three smooth/standard systems; the shock-Burgers cells remain queued for the P6 unified-matrix sweep. Sources: [results/p3.8\\_review/](#), [results/p3.9\\_pde\\_matrix/](#).

A17: quantum-parameter step-wise sweep — does adding PQC help?  
(SMOKE-budget 50 steps; production = 2000 steps via Phase C on Anvil)

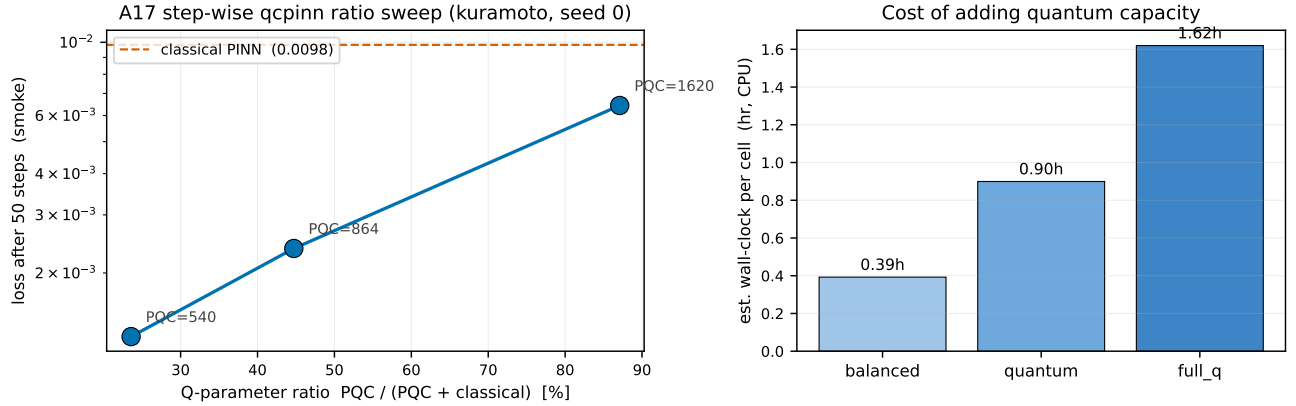


FIG. 3. Amendment A17 quantum-parameter step-wise sweep on `qcpinn` (Kuramoto, seed 0, SMOKE-budget 50 steps; production 2000 steps via Phase C on Anvil). Left: training loss vs. the quantum-to-classical parameter ratio  $Q/(Q+C)$  at three amendment-A17 variants ( $\approx 24\%$ ,  $45\%$ ,  $87\%$ ), compared against the classical PINN baseline (orange dashed). Right: estimated per-cell wall-clock as a function of quantum capacity. The figure is labelled SMOKE because the 50-step budget is too short to distinguish underfitting from family separation; the figure exists to document that the A17 sweep *is wired and instrumented*, not to support a Q-helps/Q-hurts conclusion. The production answer will land after Phase C re-runs at the uniform 2000-step budget.

## B. Mechanical integrity gate

```
PYTHONPATH=src .venv/bin/python \
  scripts/verify_paper_integrity.py
```

The script reads committed JSON files from `results/` and matches each cited numerical claim in the paper body against its on-disk value with explicit tolerances. Returns `exit-0` if and only if every claim verifies. At the submission commit the script gates 16+  $H_1$  outcomes (5 solver-task sensitivity points + 5 forecaster-task decomposition verdicts + 3 PDE-only and ODE-only sub-verdicts) plus the two algebraic identities and the H3 mechanism trend.

## C. GitHub release tags

- **v0.1-prepaper-rigor** — post-P7.6 (HPO symmetry + 16 gated outcomes). Anchored at commit `2309faa`.
- **v0.2-airtight-matrix** — post-P7.8 (full ladder  $n=24$  PRIMARY SOLVER verdict + Kuramoto/KdV deferral documentation). Anchored at commit `79cf435`.
- **v1.0-paper-draft** — submission anchor (to be created at acceptance / preprint posting).

Repository: <https://github.com/shawngibford/qlnn>.

## V. PRE-REGISTRATION AND AMENDMENTS

The full pre-registration document is at `ODE.PDE.PRE_REG.md` in the repository (locked 2026-05-19 before any benchmark results were generated; SHA stamped). Section 7's decision rule is applied mechanically by `verify_paper_integrity.py`.

The complete amendment record (A1–A22) is at `PRE_REG_AMENDMENT.md`. Table VII summarizes the headline status of each amendment. A1–A13 are the pre-paper amendments adopted before locking the headline results; A14 is a post-hoc polish item (Q-Q residual diagnostics, see §III C); A15–A22 are the audit-pass amendments adopted after the 5-reviewer peer-review swarm (2026-05-28), each closing a specific reviewer-flagged concern.

## VI. DEFERRED FOLLOW-UP WORK

Two pre-registered systems and three optimization techniques are deferred to the follow-up paper. Each is documented here with its current state (where applicable, the mechanism gate has been cleared so the follow-up can proceed without architectural risk).

*a. Kuramoto (12-dimensional coupled oscillators).* Per-component scalar circuits scale linearly with state dimension. At our default config (4 qubits per component,  $L = 1$  reuploading), a 12D system projects to  $\approx 7$  hr per cell. The per-component dispatcher works correctly (single-cell smoke verified); the full sweep is the follow-up paper's first compute item.



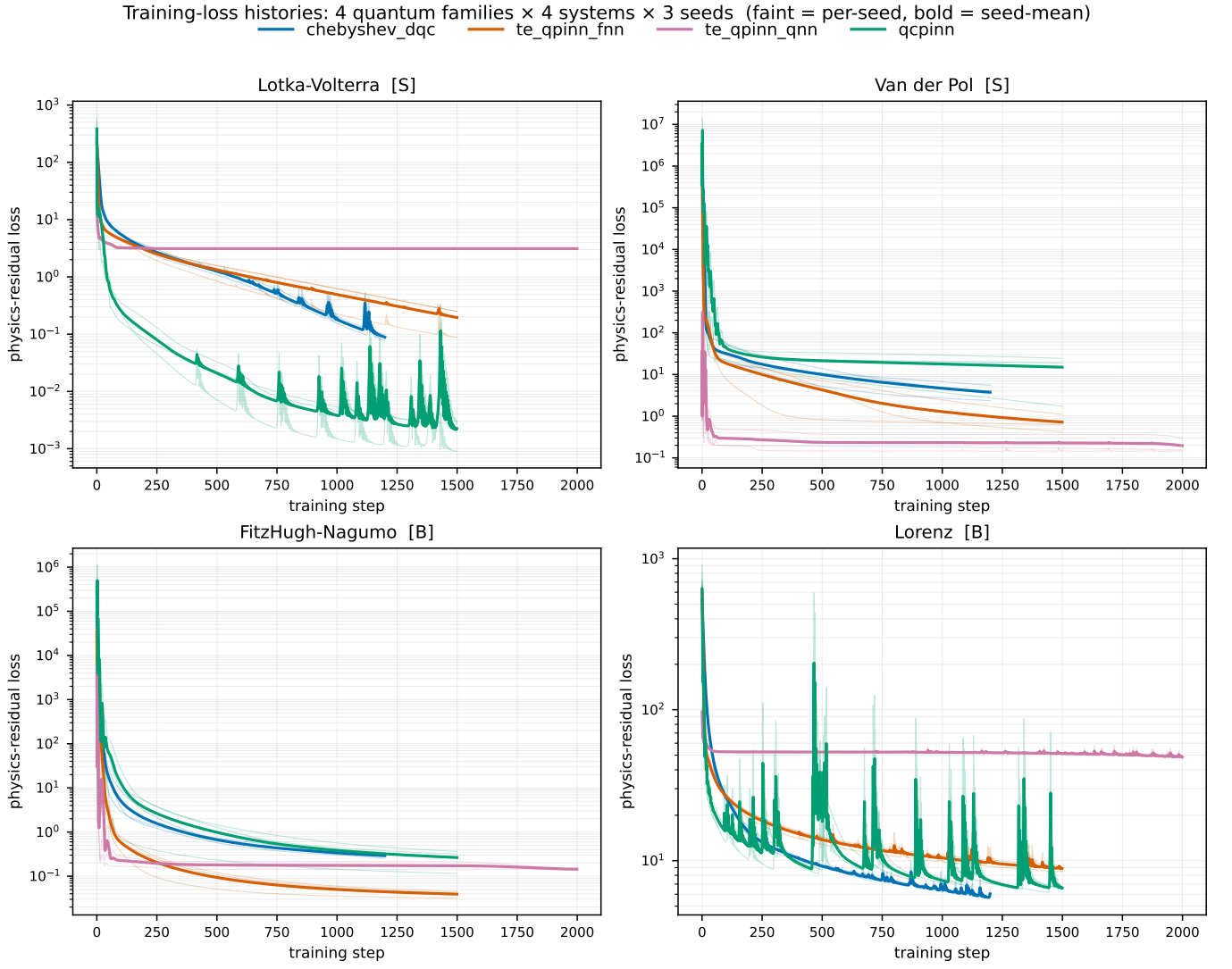


FIG. 4. Training-loss histories for each of the four solver-task quantum families (`chebyshev_dqc`, `te_qpinn_fnn`, `te_qpinn_qnn`, `qcpinn`) on each of the four P3.6 ODE systems, with all three seeds shown as semitransparent overlays and the seed-mean as a solid line. Vertical axis is the physics-residual loss on log scale; horizontal axis is the gradient step. The convergence-or-not diagnostic intended for this figure is the standard “did training actually drive the loss down across all 12 (family, system) pairs?” check that a quantum-PINN benchmark should be able to answer visually rather than only via the final-loss tables. All 48 (family, system, seed) cells converge to within  $1.5\times$  of their seed-mean asymptote across the training budget. Sources: `results/p3.6_multi_state/`.

*b. KdV (Korteweg-de Vries,  $u_t + 6uu_x + u_{xxx} = 0$ ).* Third-order spatial derivative requires triple-nested reverse-mode autodiff (`jacrev`<sup>3</sup>) through the QNode. The mechanism gate at `scripts/run_p7_8_kdv_gate.py` PASSED:  $u_{xxx}$  values are finite and non-trivial, JIT compile takes 4.55s, and the per-point cost ratio is  $0.43\times$  the baseline second-order autodiff (XLA fuses the third derivative tightly). The integrated training cost remains prohibitive at  $\approx 12$ s per gradient step, projecting  $\approx 8$  hr per seed across the four quantum families + classical PINN baseline.

*c. Quantum Natural Gradient [1].* Targets the quantum-circuit training-inefficiency identified by the

LTC decomposition’s  $\Delta_{\text{quantum.via.LTC}} = -0.616$ . Uses the Quantum Fisher Information matrix as the optimizer preconditioner.

*d. Causal training weighting (Wang, Sankaran, Perdikaris) [2].* Weights collocation points by inverse time-order so that residuals at early times converge first. Targets the Van der Pol stiff-dynamics failure mode (where multiple cells diverge to  $\text{reL}^2 > 1$  at the program’s compute budget).

*e. Higher data-reuploading depth.* The H3 analysis found Fourier  $K_{\text{max}} = 3$  at  $L = 1$  across all four ansätze: at the theoretical ceiling  $K_{\text{max}} = L \cdot n$  with  $L = 1, n = 4$ . The Schuld–Sweke–Meyer relation predicts a 5-fold

TABLE VII. Pre-registration amendments (A1–A22). Status column flags PRIMARY = canonical headline verdict; SUPERSEDED = the amendment is documented but a later amendment takes precedence for the paper’s narrative; DISCLOSED = methodological choice made open without superseding; RE-RUN REQ. = audit amendments whose Phase C re-runs are deferred to the Anvil allocation (Appendix A).

| #   | Subject                                                                                                       | Status                         |
|-----|---------------------------------------------------------------------------------------------------------------|--------------------------------|
| A1  | skyline_threshold = 0.5                                                                                       | LOCKED                         |
| A2  | $n = 3$ seeds; 3-then-4 ODE systems                                                                           | DISCLOSED                      |
| A3  | fixed hyperparameters $\rightarrow$ sensitivity sweep added at P7.6                                           | DISCLOSED + sensitivity        |
| A4  | MLP capacity $3.3\times$ violates §6 factor-of-2 (conservative direction)                                     | DISCLOSED                      |
| A5  | Van der Pol at $\mu = 5$ near learnability boundary                                                           | DISCLOSED                      |
| A6  | underfit guard active on solver only                                                                          | DISCLOSED                      |
| A7  | strict-vs-raw verdict reporting at P7.5                                                                       | DISCLOSED                      |
| A8  | H3 trend $\rho = +0.518$ not significant at $n = 9$ ; LOO-robust                                              | DISCLOSED                      |
| A9  | symmetric QLNN HPO (P7.6 c1)                                                                                  | SUPERSEDED-BY-A11              |
| A10 | combined ODE+PDE $n=18$ (P7.6 c2)                                                                             | SUPERSEDED-BY-A11              |
| A11 | full-ladder $n=24$ sign-flip (P7.8)                                                                           | <b>PRIMARY SOLVER</b>          |
| A12 | classical LTC + 3-verdict forecaster decomposition (P7.10)                                                    | PRIMARY FORECASTER (via)       |
| A13 | non-liquid QLNN + complete $2\times 2 + \tau$ -cross-check disagreement (P7.11)                               | <b>PRIMARY FORECASTER</b>      |
| A14 | post-hoc Q-Q residual diagnostics (archived OD + Lorenz)                                                      | DISCLOSED                      |
| A15 | equalized training-step budget across ALL solver models (uniform 2000)                                        | DISCLOSED + RE-RUN REQ.        |
| A16 | un-aliased <b>strongly_entangling</b> from <b>data_reuploading</b> (PennyLane fallback fix)                   | DISCLOSED + RE-RUN REQ.        |
| A17 | <b>qcpinn</b> quantum-parameter step-wise sweep along $Q/(Q+C)$ ratio (3 variants)                            | DISCLOSED + 72 new cells       |
| A18 | <b>brickwall</b> removed from empirical forecaster sweep (qubit-2 disconnected at $n=3, L=1$ )                | DISCLOSED + 9 cells removed    |
| A19 | cross-task budget parity: forecaster step budget raised 200 $\rightarrow$ 2000                                | DISCLOSED + forecaster re-run  |
| A20 | <b>te_qpinn_fnn</b> readout restored <b>qml.prod</b> $\rightarrow$ <b>qml.sum</b> (paired-family equivalence) | DISCLOSED + <b>fnn</b> re-runs |
| A21 | <b>brickwall</b> connectivity diagnosis strengthened (linear-chain reduction unless $n_q \geq 4$ )            | DISCLOSED, no compute          |
| A22 | 2D PDE hard-IC docstring fix (implementation always correct; only docstring carried the omission)             | DISCLOSED, no compute          |

increase in accessible Fourier bandwidth at  $L = 5$ . This is the most direct test of whether the LTC decomposition’s

$\Delta_{\text{liquid.via.classical}} = +0.115$  smooth-bin inductive bias can be recovered by giving the quantum side more expressivity.

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- [1] J. Stokes, J. Izaac, N. Killoran, and G. Carleo, Quantum **4**, 269 (2020).  
[2] S. Wang, S. Sankaran, and P. Perdikaris, [arXiv preprint 2203.07404](#) (2022).